

A

SYNOPSIS REPORT

on

" Integrum - AI-POWERED STUDENT SUCCESS PLATFORM"

(An Integrated Academic, Productivity and Career Management System)

Submitted in partial fulfilment of the requirements of the degree

of

BACHELOR OF SCIENCE

in

COMPUTER SCIENCE

Submitted by

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(SEM V)

(2026 – 2027)

1. Name and Title of the Project

Name : Sufiyan Usman Shaikh

Project Title: " *Integrum* - AI-Powered Student Success Platform: An Integrated Academic, Productivity and Career Management System"

Product Name (Application Brand): Integrum

2. Objective and Scope of the Project

Existing System

Students currently depend on multiple disconnected platforms to manage their academic and career-related activities — separate note-taking applications, calendar apps, placement portals, resume builders and internship-tracking websites. Because none of these tools communicate with one another, information remains fragmented across platforms, effort is often duplicated, deadlines are missed, and students have limited visibility into their overall academic and career progress.

Proposed System

The proposed Student Success Platform (Version 1.0) addresses these gaps by bringing academic management, productivity tracking and career preparation into a single, integrated web application. Instead of switching between unrelated tools, a student uses one authenticated platform in which study plans, assignments, attendance, notes, resumes, skills, internships and placement activity are all connected to the same profile and visible on a unified dashboard.

Objective

Primary Objective:

- To design and develop an integrated, AI-assisted web platform that centralises a student's academic, productivity and career-related activities within a single ecosystem, in place of multiple disconnected tools.

Secondary Objectives:

- To provide a unified dashboard through which students can track study plans, assignments, attendance and career progress.
- To embed Artificial Intelligence within existing modules, such as resume review, study planning and note summarisation, rather than as an isolated chatbot feature.
- To support the complete career-preparation journey of a student, from skill tracking to internship and placement management.
- To provide administrators with centralised control over users, academic data and platform content.
- To generate analytical insight into a student's academic performance, productivity and placement readiness.

Scope

The scope of the proposed system (Version 1.0) is limited to the following seven functional pillars, which together form a complete, self-contained academic and career management ecosystem for a single institution:

- Identity & Access, Academic Hub, Career Hub, Productivity Hub, AI Intelligence Hub, Analytics Hub and Administration Portal (detailed in Section 6).

The following are intentionally kept outside the scope of Version 1.0 and are identified as future enhancements: OCR-based document processing, a voice assistant, a dedicated recruiter portal, third-party company API integrations, a recommendation engine and a native mobile application.

3. Methodology

The project follows a structured Software Development Life Cycle (SDLC) rather than an ad-hoc, code-first approach. Development is organised into the following phases, each of which produces a concrete deliverable before implementation begins:

1. Requirement Analysis – study of the syllabus guidelines, evaluation criteria and stakeholder expectations.
2. Product Discovery – selection of the problem domain, target users and technology stack.
3. Product Definition – formulation of the problem statement, user roles and the module ecosystem.
4. Scope Planning – division of the platform into releases and freezing of the Version 1.0 module set.
5. System Design – preparation of the Software Requirements Specification, Software Architecture Document, database design (ER diagrams), UML diagrams and API design.
6. Development – implementation of the frontend, backend, database and authentication layer, carried out in sprints.
7. Testing – unit testing, integration testing and user acceptance testing of all modules.
8. Deployment – hosting of the frontend, backend and database on cloud platforms, with version control maintained on GitHub.
9. Documentation – preparation of the final report, user manual and presentation.

This phase-wise methodology ensures that architectural and design decisions are finalised before coding begins, reducing rework and keeping the final product consistent with the design.

4. Hardware and Software to be Used

Hardware Requirements

- Processor: Intel Core i5 (or equivalent) and above
- RAM: 8 GB minimum (16 GB recommended for development)
- Storage: 256 GB SSD or higher
- A stable broadband internet connection for development, testing and cloud deployment

Software Requirements

- Frontend: React, TypeScript, Tailwind CSS
- Backend: Node.js, Express.js
- Database: PostgreSQL, with Prisma as the Object-Relational Mapper (ORM)
- Authentication: JSON Web Tokens (JWT), bcrypt for password hashing
- Version Control: Git and GitHub
- Deployment Platforms: Vercel (frontend), Render (backend), Supabase / Neon (PostgreSQL hosting)
- Development Tools: Visual Studio Code, Postman, Figma (for UI/UX design)
- Operating System: Windows / Linux / macOS

Security Considerations

- User passwords will be securely hashed (using bcrypt) before being stored in the database.
- JWT-based authentication will be used to maintain secure, stateless user sessions.
- Role-based authorisation will restrict access to sensitive modules, such as the Administration Portal.
- Personal student data will be handled and stored securely, with access limited to authorised roles.
- HTTPS will be enforced across the deployed frontend and backend to secure data in transit.

5. Project Timeline

The project is planned around a target completion date in the first week of October 2026, giving approximately two and a half months for design, development and deployment.

Phase	Duration	Key Activities
Phase 1 – Planning & Design	Remaining weeks of July 2026	Finalisation of SRS, architecture, database schema, API specification and wireframes.
Phase 2 – Core Development	August 2026	Implementation of Identity & Access, Academic Hub and Career Hub modules.
Phase 3 – Feature Completion, Testing & Deployment	September 2026	Implementation of Productivity Hub, AI Intelligence Hub, Analytics Hub and Administration Portal, followed by integration testing and cloud deployment.
Phase 4 – Buffer, Documentation & Viva Preparation	1st week of October 2026	Bug fixing, final report preparation, presentation and viva-voce preparation.

6. Modules

Version 1.0 of the platform is organised into seven functional pillars. Each pillar groups a set of closely related modules, giving the system a layered and maintainable architecture.

1. Identity & Access

Handles user authentication, profile management and personalised access to the platform.

- Authentication (Login, Register, Forgot Password, Email Verification)
- Student Profile (Personal Details, Education, Skills, Career Goals, Social Links, Resume Upload)
- Dashboard (Personalised Overview, Upcoming Tasks, Quick Actions, Recent Activity)

2. Academic Hub

Enables students to manage all academic activities from a single place.

- Study Planner (Study Schedule, Daily Goals, Weekly Planner, Progress Tracking)
- Notes Manager (Upload, Subject-wise Organisation, Tags & Search, PDF/DOCX/PPT support)
- Assignment Manager (Assignment List, Deadlines, Submission Status, Priority Levels)
- Attendance Tracker (Subject-wise Attendance, Attendance %, Attendance Goals, Reports)
- Academic Calendar (Semester Calendar, Exam Schedule, Deadlines, College Events)

3. Career Hub

Supports students through their internship and placement journey.

- Resume Builder (Resume Creation, Templates, PDF Export)
- Skill Tracker (Technical Skills, Soft Skills, Certifications, Personal Projects)
- Internship Tracker (Applied Companies, Application Status, Interview Schedule, Offer Tracking)
- Placement Tracker (Company List, Eligibility Status, Placement Progress, Interview History)

4. Productivity Hub

Improves the student's personal productivity beyond academics.

- Task Manager (Personal Tasks, Academic Tasks, Priority Management, Completion Tracking)
- Reminder System (Assignment Reminders, Exam Alerts, Interview Reminders, Certification Renewal Alerts)

5. AI Intelligence Hub

Embeds Artificial Intelligence within existing workflows rather than as a standalone chatbot.

- AI Resume Review (Resume Analysis, Resume Quality Assessment, Improvement Suggestions, Skill Gap Detection)
- AI Study Planner (Personalised Timetable, Revision Planning, Topic Prioritisation, Goal Recommendations)
- AI Notes Assistant (Notes Summarisation, Key Point Extraction, Question Generation, Flashcard Generation)

AI capabilities in Version 1.0 will be implemented using modern Large Language Model (LLM) APIs, integrated directly into the Resume Review, Study Planner and Notes Assistant modules rather than exposed as a separate chatbot. The scope is limited to lightweight, assistive functions such as

summarisation, intelligent recommendations, resume evaluation and study-plan generation; advanced AI capabilities are intentionally kept outside the scope of Version 1.0.

6. Analytics Hub

Provides measurable insight into academic performance, productivity and career readiness.

- Academic Analytics (Study Progress, Attendance Percentage, Assignment Completion Rate, Subject Performance)
- Career Analytics (Resume Completion, Skills Progress, Internship Application Statistics, Placement Readiness)
- Productivity Analytics (Daily Productivity, Weekly Activity, Task Completion Rate, Goal Achievement)
- Visual Dashboard (Charts, Graphs, Progress Indicators, Performance Reports)

7. Administration Portal

Provides centralised management of the platform, its users and its academic data.

- User Management (Student Accounts, Administrator Accounts, Role Management, Account Status)
- Academic Management (Departments, Courses, Subjects, Semester Configuration)
- Content Management (Announcements, Notices, Events, Resources)
- Platform Management (Dashboard Statistics, Activity Logs, Feedback Management, System Configuration)
- Reporting (User Reports, Academic Reports, Usage Analytics, Platform Health Reports)

7. Application and Future Scope of the Project

Applications

- Provides students with a single platform to manage academics, productivity and career preparation, eliminating dependence on multiple disconnected applications.
- Helps reduce missed deadlines and duplicated effort through consolidated tracking of assignments, attendance and study plans.
- Assists students in career readiness through resume building, skill tracking and placement/internship tracking, all AI-supported.
- Gives institutions a centralised administration portal to manage students, academic data and platform-wide announcements.
- Offers measurable analytics that help students and institutions monitor academic and career progress over time.

Future Scope

Beyond Version 1.0, the platform can be extended with a recruiter portal, faculty dashboard, OCR-based note digitisation, a voice assistant, a recommendation engine, direct company/API integrations and a native mobile application. These enhancements are planned as part of a future Version 2.0 and are outside the scope of the present submission.

8. References

1. Sommerville, I., Software Engineering, 10th Edition, Pearson Education.
2. React Documentation – <https://react.dev>
3. Node.js Documentation – <https://nodejs.org/en/docs>
4. PostgreSQL Documentation – <https://www.postgresql.org/docs>
5. Prisma ORM Documentation – <https://www.prisma.io/docs>
6. IEEE research papers on integrated academic and career management platforms for higher education (IEEE Xplore Digital Library).